

Every essential arrangement $\mathcal{A}$ of n hyperplanes in $\mathbb{R}^d$ has at least $2n$ simplicial regions.

More precisely, adjacent to every hyperplane there are at least $2d$ simplicial regions, and nonadjacent to any given hyperplane there are at least $2(n - d)$ simplicial regions.

(Note that the simplicial regions come in pairs, since the opposite of a simplicial cone is simplicial as well). From Shannon's theorem, we derive the following theorem about zonotopes.

Theorem 7.20 (Shannon's theorem for zonotopes).

Every d -zonotope with n zones has at least $2n$ simple vertices.

More precisely, on every zone there are at least $2d$ simple vertices, and disjoint from any given zone there are at least $2(n - d)$ simple vertices.

The proof for Shannon's theorem for arrangements is not difficult: we refer to Roudneff & Sturmfels [467], where several different proofs are presented, and to [96, Thm. 2.1.5]. The interesting thing is that there is no "entirely combinatorial" proof. In fact, the corresponding statement for oriented matroids is false. As we will see below, this translates into very interesting effects for "zonotopal tilings."

7.4 Nonrealizable Oriented Matroids

We have by now *seen* so many oriented matroids around that you shouldn't be scared any more if you hit their axiomatic definition a few lines down. In fact, the axioms below just describe abstractly the "most important" properties shared by the sign vector systems of — equivalently —

- hyperplane arrangements
- zonotopes
- vector configurations
- affine point configurations (vertices of polytopes!).

We have seen that in each of these cases we get a sign vector system of the form $\mathcal{V}^* = \text{SIGN}(U)$, for a linear subspace $U \subseteq \mathbb{R}^n$.

For the following, choose $\mathcal{V}^*$ to arise from any one of the above models: perhaps it is best to take $\mathcal{A} = \{H_1, \dots, H_n\}$ to be an oriented, essential arrangement of hyperplanes, and let $\mathcal{V}^*$ be the set of sign vectors of all its cones. (For example, take the one sketched in the proof of Theorem 7.16.)

We need the following operations on sign vectors. The *zero vector* $\mathbf{0}$ and the *negative* $-\mathbf{u}$ of a sign vector $\mathbf{u}$ have their obvious meanings. The *support*

of a sign vector U is $\text{supp}(U) := \{i : u_i \neq 0\}$. The *composition* of two vectors U, V is defined componentwise by

$$(U \circ V)_i := \begin{cases} u_i & \text{if } u_i \neq 0, \\ v_i & \text{otherwise.} \end{cases}$$

The *separation set* for U, V is defined by

$$S(U, V) := \{i : u_i = -v_i \neq 0\}.$$

Finally, if $j \in S(U, V)$, we say that w *eliminates* j *between* U and V if

$$w_j = 0 \quad \text{and} \quad w_i = (U \circ V)_i \quad \text{for all } i \notin S(U, V).$$

This may look like a lot of definitions, but they all have very concrete meanings for (sets) of sign vectors, and a concrete geometric interpretation, as follows.

Definition 7.21 (Oriented matroids).

A collection $\mathcal{V}^* \subseteq \{+, -, 0\}^n$ is the set of *covectors of an oriented matroid* if it satisfies the following *covector axioms*:

- (V0) $\mathbf{0} \in \mathcal{V}^*$
("The zero covector is always a covector")
- (V1) $U \in \mathcal{V}^* \implies -U \in \mathcal{V}^*$
("The negative of a covector is always a covector")
- (V2) $U, V \in \mathcal{V}^* \implies U \circ V \in \mathcal{V}^*$
("The set of covectors is closed under composition")
- (V3) $U, V \in \mathcal{V}^*, j \in S(U, V) \implies \exists w \in \mathcal{V}^* : w \text{ eliminates } j \text{ between } U \text{ and } V$
("The set of covectors admits elimination")

The *rank* of the oriented matroid $\mathcal{V}^*$ is the largest number r such that $\mathcal{V}^*$ contains a chain of covectors of length r :

$$\mathbf{0} < X^1 < X^2 < \cdots < X^r \quad \text{with } X^i \in \mathcal{V}^*.$$

In this case, we write $r(\mathcal{V}^*) = r$ for the rank of $\mathcal{V}^*$.

Proposition 7.22. *Consider a linear subspace $U \subseteq \mathbb{R}^n$ of dimension r . Then the set of column sign vectors*

$$\text{SIGN}(U) = \{\text{sign}(\mathbf{u}) : \mathbf{u} \in U\} \subseteq \{+, -, 0\}^n$$

is the covector set of an oriented matroid of rank r .

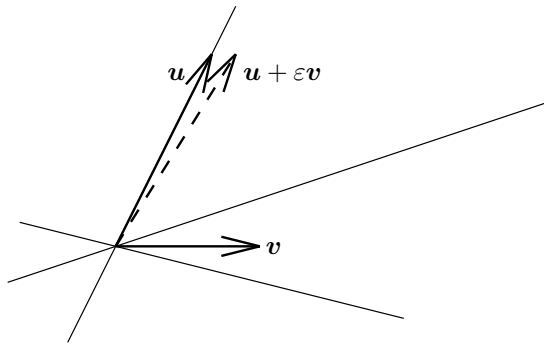
(This proves that the realizable oriented matroids $\mathcal{M}(V)$, as given by Definition 6.5, are indeed oriented matroids of rank r in the sense of Definition 7.21, if we use $U := \text{Val}(V)$ to derive the covector system $\mathcal{V}^*$.)

Proof. This is easy, but *important*, because it provides the “geometric meaning” of the axioms and of the operations on sign vectors.

The hyperplanes $\{x_i = 0\}$ induce an essential hyperplane arrangement in U , via $H_i := \{\mathbf{x} \in U : x_i = 0\}$. For these hyperplanes within U , positive sides are uniquely determined via $H_i^+ = \{\mathbf{x} \in U : x_i \geq 0\}$. With this construction, we see that $(\mathcal{V}^*, \leq)$ is the face poset of the essential, oriented hyperplane arrangement $\mathcal{H} = \{H_1, \dots, H_n\}$ in U . Thus this poset has length r .

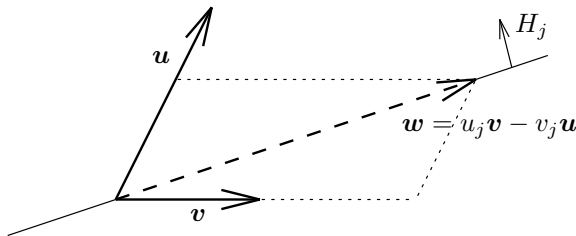
For axiom (V0), we note that $\mathbf{0} \in U$ has the sign $\mathbf{0} = \text{sign}(\mathbf{0}) \in \text{SIGN}(U)$. For (V1), take $\mathbf{u} = \text{sign}(\mathbf{u})$ for some $\mathbf{u} \in U$. From $\mathbf{u} \in U \implies -\mathbf{u} \in U$ we get $-\mathbf{u} = -\text{sign}(\mathbf{u}) = \text{sign}(-\mathbf{u}) \in \text{SIGN}(U)$.

The idea for the composition operation in (V2) is that if $\mathbf{u}, \mathbf{v} \in U$, then for any $\varepsilon \in \mathbb{R}$ we have $\mathbf{u} + \varepsilon\mathbf{v} \in U$.



Now, for any $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, if $\varepsilon > 0$ is small enough, then we get $\text{sign}(\mathbf{u}) \circ \text{sign}(\mathbf{v}) = \text{sign}(\mathbf{u} + \varepsilon\mathbf{v})$ — this is easy to see by looking at the sign vectors componentwise. From this we obtain $\text{sign}(\mathbf{u}) \circ \text{sign}(\mathbf{v}) \in \text{SIGN}(U)$.

For elimination as in (V3), we use that with $\mathbf{u}, \mathbf{v} \in U$, general linear combinations of $\mathbf{u}$ and $\mathbf{v}$ are contained in U , in particular the combination $\mathbf{w} := u_j\mathbf{v} - v_j\mathbf{u}$. Now if $u_j > 0$ and $v_j < 0$, then this is a positive linear combination of the two vectors $\mathbf{u}$ and $\mathbf{v}$.



Considering the individual components of the sign vectors, we see that the j th coordinate of $\mathbf{w}$ is zero. Similarly, if the i th coordinates of $\mathbf{u}$ and $\mathbf{v}$ don't have opposite sign, then the i th coordinate of any positive linear combination has the sign $\text{sign}(w_i) = (\mathbf{u} \circ \mathbf{v})_i$, as required. $\square$

The axioms of Definition 7.21 provide an entirely combinatorial model for the geometry of hyperplane arrangements, vector configurations, point configurations, or zonotopes. There are two (closely related) questions that we will not avoid here:

- How good is this model — how closely do the combinatorics of oriented matroids represent a situation of an actual geometric object in real space?
- What is this model good for?

In the following, we try to answer both questions.

Remark 7.23: “How good is the oriented matroid model?”

- The model is **excellent**. All the basic structural properties that we have proved in the realizable case in Section 6.3 extend to the case of general oriented matroids. In particular, we have
 - duality (as in Definition 6.10)
 - equivalence of various types of data (as in Corollary 6.9)
 - deletion and contraction as basic operations (as in Proposition 6.11).
- The *topological representation theorem* of Lawrence [207] [96, Ch. 5] shows that every oriented matroid is “nearly” realizable: it may not correspond to a real hyperplane arrangement, but it does correspond to an *arrangement of pseudohyperplanes*, which need not be straight but may be topologically deformed in some mild way.

The rest of this section will sketch this in the case of $r = 3$, which corresponds to “arrangements of pseudolines” in the plane, as investigated by Grünbaum [255]; see also [96, Ch. 6].

- For certain ranges of parameters, in particular for $r \leq 2$, for $r \geq n - 2$, and for $n \leq 7$, every oriented matroid is realizable, so the model is perfect.
- Even the nonrealizable oriented matroids come up “in practice.” We will present one example for this in the next section, where nonrealizable oriented matroids appear in the study of zonotopal tilings.

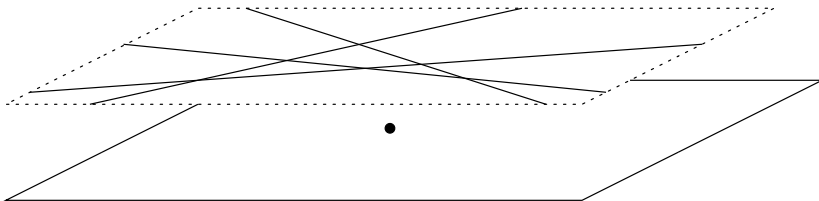
Remark 7.24: “What are oriented matroids good for?”

- Oriented matroids explain constructions like Gale diagrams and the Lawrence construction (Lecture 6), which are clearly useful and have various applications.

- Oriented matroids provide a unifying framework, consistent terminology, and widely applicable tools for several areas of geometry.
- Within the last twenty years, an extensive theory for oriented matroids was developed, with many nontrivial results. Results about oriented matroids are easily transported from one field of applications (e.g., hyperplane arrangements) to another one (like polytopes). For the existing body of theory, we refer to the book by Björner et al. [96], and the handbook chapter by Bokowski [112].
- The theory of oriented matroids allows us to handle, in a precise sense, the combinatorics of objects that are geometric (like certain simplicial spheres) but that cannot be represented in real “Euclidean space” (or at least not as polytopes).
- Thus, oriented matroids appear as a natural intermediate step in the classification of (simplicial) spheres into polytopes and nonpolytopes: this approach was pioneered by Bokowski; see [20] [16] [117] and the monograph by Bokowski & Sturmfels [118].

We will now start to discuss one topic where nonrealizable oriented matroids come up — the study of pseudoline arrangements. To make the connection, consider an arrangement $\mathcal{A}$ of n hyperplanes in $\mathbb{R}^d$ (all hyperplanes through the origin!). To draw and represent this, we consider the intersection with an affine hyperplane: a hyperplane not through the origin, but parallel to one of the hyperplanes in the arrangement. Thus we get an *affine hyperplane arrangement* $\mathcal{A}_{\text{aff}}$, consisting of $n - 1$ affine hyperplanes. (Thus, an *affine hyperplane* need not contain the origin $\mathbf{0}$, but if we talk only about a *hyperplane*, then we mean a *linear hyperplane*, which contains the origin.)

The entire geometric combinatorial structure of the hyperplane arrangement $\mathcal{A}$ can easily be reconstructed from its “affine picture” $\mathcal{A}_{\text{aff}}$, up to linear isomorphism, so nothing is lost by this reduction to affine space. In particular, one can read off the oriented matroid of $\mathcal{A}$ from the arrangement $\mathcal{A}_{\text{aff}}$, assuming that the hyperplanes in $\mathcal{A}_{\text{aff}}$ are labeled and that positive sides are determined.

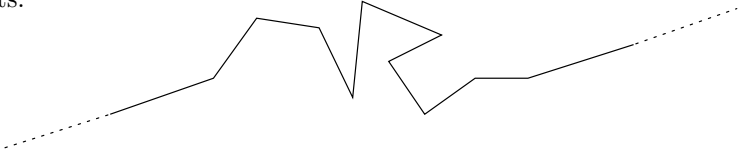


So in our drawing, for the case $d = 3$, we obtain an affine arrangement $\mathcal{A}_{\text{aff}}$ of $n - 1 = 4$ lines in the affine (dotted) plane, from an original arrangement

of 5 hyperplanes, namely the horizontal one, and the 4 planes determined as the affine hull of the 4 lines together with the origin.

This is how — in the case $d = 3$ — affine arrangements of lines in $\mathbb{R}^2$ represent 3-dimensional (hyper)plane arrangements. Every arrangement of lines in the plane determines an oriented matroid of rank 3.

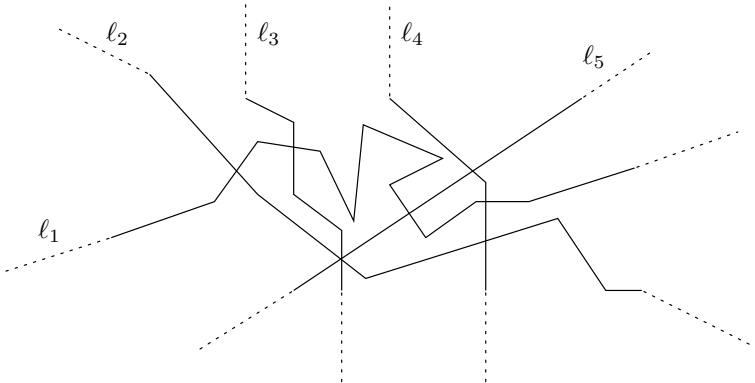
However, it turns out that if we have an arrangement of “nonstraight” lines in the plane, then we can also, in the same way, read off an oriented matroid of rank 3. There are various — quite general — ways to define such “nonstraight” lines. Basically, any type of two-way unbounded topological curves will do. See Grünbaum [255] or [96, Ch. 6] for general versions. Here, for simplicity, we will use a simpler version, with the same combinatorial results.



Definition 7.25. A *pseudoline* is a polygonal curve without self-intersections, with finitely many break points in $\mathbb{R}^2$, and whose ends “head off to infinity” in opposite directions.

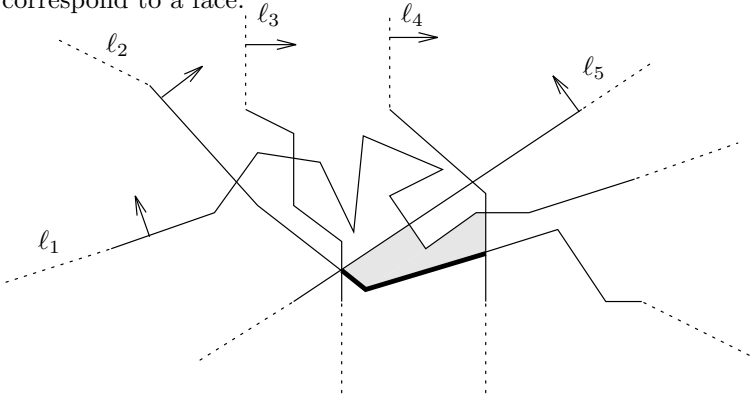
An *arrangement of pseudolines* is a finite set of pseudolines in the plane such that

- (i) any two pseudolines either are disjoint (then we call them *parallel*), or they meet in a single point and cross in this point, and
- (ii) being parallel is transitive (that is, if a pseudoline intersects one pseudoline of a parallel pair, then it also has to intersect the other one).

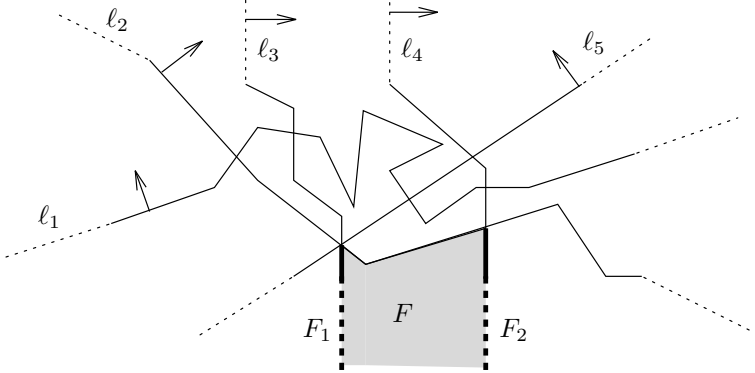


Now let $\mathcal{P}$ be an arrangement of $n - 1$ pseudolines, which have been labeled $\ell_1, \dots, \ell_{n-1}$, and where a positive side has been chosen for each of them. For every $X \in \{+, -, 0\}^{n-1}$, let F^X be the set of all those points $\mathbf{x} \in \mathbb{R}^2$ which lie on the positive side of ℓ_i if $X_i^F = +$, on the negative side of ℓ_i if $X_i^F = -$, and on the pseudoline ℓ_i if $X_i^F = 0$. This set F^X may be empty: if not, it is called the *face* of $\mathcal{P}$ associated with X .

For example, in our next drawing small arrows are used to indicate the positive side for each pseudoline. The shaded region (without its boundary) is the face associated with $X = (-++--)$, the bold edge (without endpoints) is the face associated with $(-0+--)$, whereas $(-+0--)$ does not correspond to a face.



Similarly, we assign labels to the *faces at infinity*: namely, we get a “face at infinity” for every unbounded face in $\mathcal{P}$, where we have to take into account that “parallel lines meet at infinity.” Thus, in the following drawing, the two bold edges F_1, F_2 and the shaded unbounded face F , with $X^{F_1} = (--0--)$, $X^{F_2} = (--+0-)$, and $X^F = (--+--)$, all determine the same face G^Y at infinity, with $Y = (--00-)$.

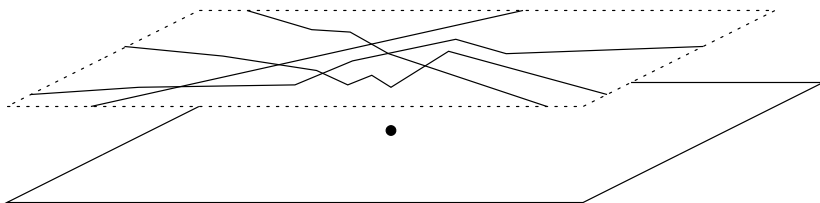


Theorem 7.26. *Let $\mathcal{P} = \{\ell_1, \dots, \ell_{n-1}\}$ be a labeled arrangement of pseudolines, for which positive sides have been chosen. Then the family of sign vectors*

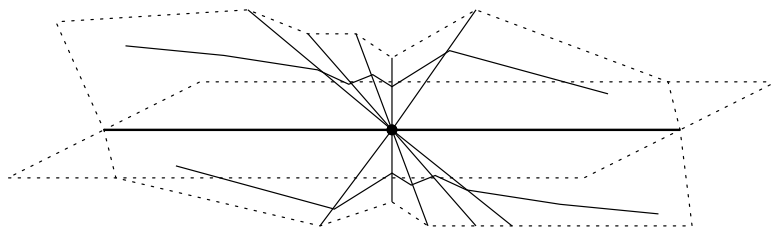
$$\begin{aligned} \mathcal{V}^*(\mathcal{P}) &:= \{ (X^F, +) : F \text{ is a face of } \mathcal{P} \} \\ &\cup \{ (Y^G, 0) : G \text{ is a face at infinity for } \mathcal{P} \} \cup \{(\mathbf{0}, 0)\} \\ &\cup \{ (-X^F, -) : F \text{ is a face of } \mathcal{P} \} \subseteq \{+, -, 0\}^n \end{aligned}$$

is an oriented matroid of rank 3.

Proof. From an arrangement $\mathcal{P}$ of $n - 1$ pseudolines, the previous procedure reconstructs a linear arrangement of n “pseudoplanes” through the origin in $\mathbb{R}^3$, where the n th pseudoplane is a straight plane, corresponding to the line at infinity in $\mathcal{P}$.



The following drawing shows two pseudoplanes of the resulting arrangement in $\mathbb{R}^3$: the flat one corresponds to the line at infinity for $\mathcal{P}$, and the nonflat one corresponds to a pseudoline in $\mathcal{P}$.



The oriented matroid $\mathcal{V}^*(\mathcal{P}) \subseteq \{+, -, 0\}^n$ arises from this arrangement in the same way as in the case of a straight arrangement of planes in $\mathbb{R}^3$.

Similarly, the proof that the system $\mathcal{V}^*(\mathcal{P})$ is an oriented matroid is analogous to the realizable case in Proposition 7.22 — except that the “linear algebra arguments” of that proof have to be replaced by “combinatorial arguments” that remain valid in the nonlinear case.

We leave the details to the reader, and refer to [96, Sect. 5.1], where the proof is nicely done in even greater generality. $\square$

In fact, there is a surprisingly strong theorem available here: Lawrence’s *topological representation theorem* states that

every linear arrangement of n pseudohyperplanes in $\mathbb{R}^d$ yields an oriented matroid $\mathcal{V}^* \subseteq \{+, -, 0\}^n$ of rank d .

Theorem 7.26 just presents the case $d = 3$ of this statement. (We do not intend to give a precise definition of arrangements of pseudohyperplanes here. Intuitively this should be clear; see [96, Ch. 5] for a careful explanation.) The topological representation theorem, however, also includes a converse:

every oriented matroid on n elements of rank d can be represented by an arrangement of n linear pseudohyperplanes in $\mathbb{R}^d$, which is essentially unique.

Thus we have a bijection

oriented matroids $\longleftrightarrow$ (equivalence classes of) pseudoarrangements.

The second half of the theorem, constructing a pseudoarrangement for a given oriented matroid, is by far the harder part to prove: it is not easy even in the special case of $d = 3$.

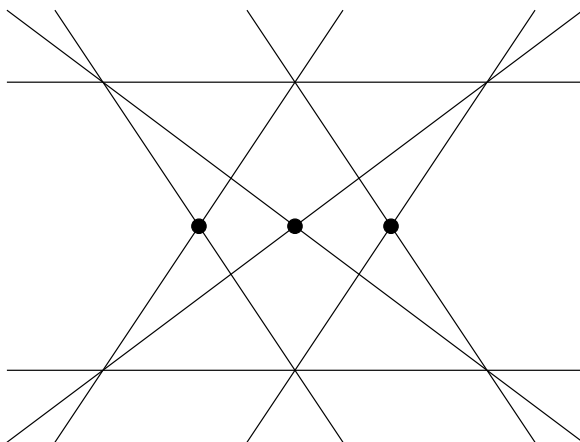
Complete proofs of the topological representation theorem have been given by Folkman & Lawrence [207], by Edmonds & Mandel [191] (who first proved the stronger piecewise linear version corresponding to our version of polygonal pseudolines), in Bachem & Kern [35], and in Björner et al. [96, Chs. 4 and 5].

Definition 7.27. Two arrangements of pseudolines are *combinatorially equivalent* if — possibly after relabeling, and after change of the positive sides — they have the same oriented matroid.

An arrangement of pseudolines $\mathcal{P}$ is *realizable* (or *stretchable*) if it is combinatorially equivalent to a (straight) arrangement of lines, that is, if the oriented matroid $\mathcal{V}^*(\mathcal{P})$ is realizable.

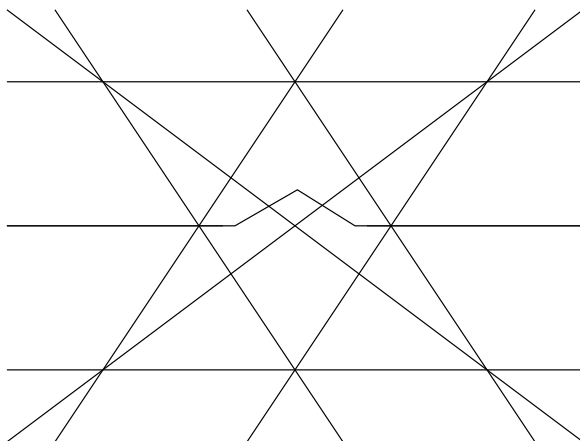
Are pseudoline arrangements really more general than line arrangements? Or is every pseudoline arrangement stretchable? Well, nonrealizable pseudoline arrangements do exist, and they are not even that hard to construct. The following construction of a nonrealizable pseudoline arrangement already appeared in Levi's 1926 paper [362], the first paper on pseudoline arrangements ever written (as far as I know).

Example 7.28. (Levi [362]) Consider the following arrangement of 8 lines in the plane, the well-known *Pappus configuration*.



Pappus' theorem states that the three black dots are collinear in every line arrangement that is combinatorially equivalent to this arrangement. This implies that there is no straight representation of the following pseudoline

arrangement, the *non-Pappus configuration*.



However, this pseudoline arrangement *does* give an oriented matroid via Theorem 7.26 (uniquely, if we number the pseudolines and specify a positive side for each of them), the *non-Pappus oriented matroid* $\mathcal{M}_{nP}$. This is an oriented matroid of rank 3 on 10 elements. However, if we delete the element that corresponds to the line at infinity, then we get an oriented matroid $\mathcal{M}'_{nP}$ on 9 elements which is nonrealizable as well, since the line at infinity was not needed for the nonrealizability argument.

There are even nonrealizable pseudoline arrangements that are simple (no three pseudolines cross in a point or are parallel) — see Ringel [463] [255, p. 42] and Exercise 7.16 for an example with only 9 pseudolines in the plane. In fact, all (simple or nonsimple) arrangements of pseudolines with at most 8 pseudolines are realizable (for this count the line at infinity, if it is there), and Ringel’s example is essentially the unique simple one with 9 pseudolines, according to Richter [455].

7.5 Zonotopal Tilings

What do you “see” if you “look at” a d -dimensional zonotope? You see its “front facets,” which are $(d - 1)$ -dimensional facets and which fill a shape that is a projection of a zonotope, and thus is a zonotope itself.

If $d = 3$ (where this is most likely to happen to you anyway), and you look from a point very far away, the picture might look like the drawing on the next page. Here the shape you see is a 10-gon, filled “face to face” by quadrilaterals and hexagons.

In general, the shape you see is a (parallel) projection of a zonotope, which is a zonotope itself. It is covered by the images of the front facets of the zonotope, which are also zonotopes.